IETF 127 · NETMOD

Validating anydata in YANG Library context

draft-netana-netmod-yang-anydata-validation-00

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Replaces draft-ietf-netmod-yang-anydata-validation-00 (DEAD), as requested by the chairs on 27 April 2026

16.11.2026

STATUS

How we got here

Nov–Dec 2025

IETF 124 poll shows good support for adoption. -00 posted under the draft-ietf name. List review by Rob Wilton and Andy Bierman.

Mar 2026

James Cumming proposes a “validate” substatement. IETF 125 hackathon validates YANG-Push at a message broker.

Apr 2026

Post factum adoption call and WGLC threads: Rob, Andy, Tom, Holger, Reshad, Jeroen. On 27 April the chairs asked for a new individual I-D that addresses the points; the WG document went to DEAD.

Oct 2026

draft-netana-netmod-yang-anydata-validation-00, dated 29 October. Informational. Rewritten to answer the adoption call.

Goal for this session: confirm the comments are addressed and issues are fixed, settle where a new YANG statement extension belongs.

Threads: adoption call · WGLC · new draft announced 30 Oct

MOTIVATION

Wrong data passes through, and nobody checks it

anydata output is, by default, not checked against any schema on the receiving side: YANG-Push datastore-contents, compare results, Patch edits.

Consumers sit behind message brokers, far from the producer, with no session to learn its schema. A defect in the network OS reaches billing, capacity planning and dashboards.

Schema validation is the first gate: it catches what the model forbids, such as a negative counter64, an unknown node or a missing key. Plausibility checks come after it.

> “The data processing chain accepted the value and the dashboards went nuts…”
>
> Holger Keller (DT), on a device reporting 80 million Watts for a few seconds each quarter, filtered out after manual investigation. netmod, 20 Apr 2026

WHAT CHANGED SINCE -00

From a two-paragraph method to a defined procedure

Topic	draft-ietf-...-00 (Nov 2025)	draft-netana-...-00 (Oct 2026)
Status	Standards Track	Informational
Scope	Use YANG Library (RFC 8525) to validate anydata	Defines the validation context; YANG Library is one way to obtain the schema context (§1)
Survey	Two groups, seven documents	21 anydata nodes in 12 RFCs and drafts, with anchor, datastore and fragment for each (§3)
Schema lookup	One bullet per encoding	Schema anchor, lookup procedure, first-level rules for XML, JSON and CBOR, nested anydata and mount points, outcomes (§4)
Datastore	Not an input	An explicit input; rules follow the source datastore (RFC 8342), no per-datastore default (§1, §5)
Validation	One paragraph	anydata-candidate and anydata-complete defined against RFC 7950 §8.1, a procedure, and error-tag mapping (§5)
Example	None	YANG-Push counter with a negative value (§5.4)

CORE IDEA

A validation context per anydata instance

1 · Schema context

Modules, revisions, features, deviations. YANG Library is one source.

2 · anydata node

Which node is validated. One message can carry several.

3 · Schema anchor

Parent of the first-level nodes, from the message: Patch target, data-path, path-prefix. Root “/” otherwise.

4 · Datastore

Which schema and rules apply, e.g. no config false in running; RFC 8342 §5.3 for operational.

anydata-candidate

RFC 7950 §8.1 “true in all data trees”: types, ranges, patterns, keys, one case per choice, if-feature, config. No must, when, unique, mandatory, min/max-elements; references need not resolve. For fragments.

anydata-complete

Adds every “valid data tree” property, on the content plus the ancestors built from the anchor. For complete content. Extra datastore data may be used to resolve outward references.

The consumer picks the option per instance; no default. Missing context gives “not validated”, never “invalid”.

Adoption call and WGLC threads

Who	Comment	Resolution	§	Status
Chairs	New I-D, address both threads	New individual draft, this version	–	Done
Holger	RESTCONT, propeses, "The element's", avaiable	Text rewritten	1, 4, 6	Done
Reshad	Too sparse; show how the counter is caught	Lookup, validation, worked example	4, 5.4	Done
Reshad	Is strict parsing the only libyang change?	No: also a candidate flag and subtree-only checks	6	Open
Rob	One or several anchors?	One anchor per anydata instance	4.1, 4.4	Done
Rob	Rules should follow the datastore	Datastore is an input; consumer picks option for fragments	1, 5.1	Discuss
Rob	Informational; unclear what it enforces	Informational; procedure and error table	5.2, 5.3	Done
James	How is validation opted in or out? (IETF 124)	Consumer decides; anydata statement unchanged	1	Done
James	Do not tie it to YANG Library	Abstract and §1 changed; title not yet	1	Partly
James	"validate" substatement in anydata	Not in this version; question 2 for the WG	1	WG

Mails: Chairs 27 Apr · Holger 20 Apr · Reshad 20 Apr · Rob 17 Apr · James 19 Mar

Rob, Andy, Jeroen, Thomas, Tom

Who	Comment	Resolution	§	Status
Rob	Not always from the root: Patch, 9144, Push v2	Schema anchor, taken from the message	4.1, 4.4	Done
Rob	RFC 9144 output is relative to the filter	A filter does not change the anchor; reason given	4.4	Done
Rob	Any datastore; RFC 8342 §5.3 for operational	Datastore is the fourth input	1, 5.1	Done
Rob	Pass instance data to resolve leafrefs	Allowed; how it is obtained is out of scope	5.1	Done
Rob	Do not constrain every anydata	anydata unchanged, validation optional	1	Done
Andy	What problem is solved?	Decoupled consumers, counter example	1, 5.4	Done
Andy	Follow the RFC that defines the node	Survey of 21 uses, anchor per use	3, 4.4	Done
Andy	Use standard annotations	Not needed; reason stated	4.5	Done
Jeroen	Scope; no module names; unknown module	Receiver-side context; error or not validated	4.3, 4.6	Done
Thomas	Cite RFC 7951 §5.5, RFC 9254 §4.5	Cited in the encoding rules	4.3	Done
Tom	Use BCP 14 keywords in the rules	Rules rewritten with MUST, SHOULD	4, 5	Done

Mails: Rob Dec · Andy Dec · Rob 7 Apr · Andy 7 Apr · Tom 8 Apr · Thomas 19 Apr · Jeroen 20 Apr

STILL OPEN · PLANNED FOR -01

What this version does not yet fix

Security Considerations

Still TBD (§8). Text drafted: untrusted content, bounded XPath evaluation, trusted schema context.

Implementation Status

Only root-anchored strict parsing (§6); LYD_PARSE_ANYDATA_STRICT is now in libyang devel. Rewrite per RFC 7942.

Nested notifications

“First-level node MUST be a notification” rejects notifications defined in a container or list (RFC 7950 §7.16.2). Fix in §4.1 and Table 1.

Notifications under anydata-complete

Leafrefs from a notification into the datastore (RFC 7950 §6.4.1) are not in the content. Report them as not validated.

Title

Still “in YANG Library context”, which James asked to drop. Proposed: “Validating YANG anydata Content”.

Datastore alone or consumer choice

Rob’s 17 April view versus fragment content. For discussion here.

EXAMPLE (§5.4)

A negative counter in a YANG-Push update

```
<push-update xmlns="...:ietf-yang-push">
 <id>1011</id>
 <datastore-contents>
  <interfaces xmlns="...:ietf-interfaces">
   <interface>
    <name>eth0</name>
    <statistics>
     <in-octets>-4271233</in-octets>
    </statistics>
   </interface>
  </interfaces>
 </datastore-contents>
</push-update>
```

Context

Schema context of the publisher · anydata node datastore-contents · anchor "/" · datastore <operational> · option anydata-candidate

Result: invalid

in-octets is yang:counter64 (0 to 2^64-1). error-tag invalid-value at /if:interfaces/if:interface[if:name='eth0']/if:statistics/if:in-octets

Why not anydata-complete

The update is a fragment, so it would also report the missing mandatory "type" leaf, although the content is a correct part of <operational>.

The draft defines the anchor; an extension can say where it is

This document: what

The schema anchor of each anydata instance, how it is resolved, and how first-level nodes are looked up (§4). Today the anchor is found per use: Patch target, data-path, path-prefix, root.

Extension: how it is conveyed

A YANG extension module, yang-anydata-ext, that a module puts on its anydata node so a library finds the anchor and the mode without code per message type.

```
anydata value {
  ext:schema-anchor "../target";
  ext:validation-mode anydata-candidate;
}
```

Sketch only, not in the draft. The anchor stays a run-time value: each edit has its own target, so the statement points to where the value travels, a sibling leaf or an RFC 7952 annotation on the anydata node.

Open: in this document (then Proposed Standard) or a separate one, and how existing modules adopt it.

QUESTIONS FOR THE WORKING GROUP

Next steps

1. Are the comments from the list addressed?
   Including the open items on the previous slide, which we plan to close in -01.

2. Where does a YANG statement for anydata go?
   An anydata substatement or extension carrying the validation mode, as James proposed, and the anchor. In this document, which then becomes Proposed Standard, or in a separate document, keeping this one Informational?

3. Please review and give us feedback.